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[Advanced Autonomous Systems Integrator] | Capability Statement | June 2026 **CIEMS + AI Mechanic: Unified Autonomy Assurance for Tactical UAS** Group 3-4 UAS | Runtime AI Integrity | Lifecycle Configuration Management | Forensic Accountability

ADMINISTRATIVE IDENTIFIERS

CAGE Code: [CAGE-XXXXXX]	UEI: [UEI-XXXXXXXXXX]
NAICS: 541715 541511 334511	PSC: AC14 7R
Contract Vehicles: GSA MAS IT Schedule 70; SEWP V; DIU OTA; AFWERX Phase II	Small Business Status: [Small Business / 8(a) / SDVOSB]
Facility Clearance: Secret; Personnel TS/SCI available	Point of Contact: [Program Manager Name], [email@company.com], [phone]

CORE CAPABILITY

We deliver runtime autonomy assurance and lifecycle integrity for AI-governed UAS platforms — closing the accountability gap between algorithm performance and operational trust. The **Configurable Integrated Engineering Management System (CIEMS)** and **AI Mechanic** constitute a unified, end-to-end solution purpose-built for Group 3-4 tactical UAS operating under autonomous or semi-autonomous mission authority.

AI Mechanic provides onboard, real-time behavioral health monitoring of AI inference engines during autonomous operations. It detects behavioral drift, adversarial sensor manipulation, and confidence score anomalies within **<200ms**, triggering a graduated **Autonomy Control Engine (ACE)** intervention response that preserves mission continuity while restoring human oversight. All AI inference events are written to a

tamper-evident forensic record enabling post-incident reconstruction in hours, not weeks.

CIEMS enforces end-to-end software supply chain integrity from model training through live operational deployment. Cryptographically-signed model provenance, **Software/Hardware Bill of Materials (SBOM/HSBOM)** verification, and automated pre-flight **Mission Readiness Reports (MRR)** satisfy airworthiness documentation requirements and directly support coordination with Army and Air Force airworthiness authorities. The unified architecture aligns with **DoDD 3000.09**, DoD AI Ethics Principles, **NIST AI RMF**, and **MIL-STD-882E**.

Key Differentiators Only integrated solution combining onboard runtime AI monitoring (AI Mechanic) with full engineering lifecycle integrity management (CIEMS) in a single, coherent architecture — no integration seams, no accountability gaps. Platform-agnostic: integrates with Group 3-4 UAS without airframe modification via the standardized CIEMS Integration Module (CIM). Forensic Replay Module provides sub-second temporal resolution of all AI inference events, enabling complete post-incident mission reconstruction. Drift Detection Engine employs CUSUM/EWMA statistical process control — mission-proven anomaly detection methods adapted for AI behavioral monitoring across flight phases. Autonomy State Bus (ASB) supports STANAG 4586, MIL-STD-1553, and ROS2/DDS — compatible with existing DoD UAS ground and onboard data infrastructure. Graduated intervention architecture (alert → watchdog → graceful degradation → safe-state handover) maintains human operator authority without forcing mission abort. Designed for IL4/IL5 classified cloud deployment with a DISA STIG-aligned cybersecurity posture.

Verified Performance Thresholds

>97% Anomaly Detection Rate	<2% False Positive Rate
<200ms ACE Detection-to-Alert Latency	<500ms GCS Operator Notification
<2s Safe-State Handover Execution	<90s Pre-Flight MRR Generation
100% SBOM Coverage (pre-mission)	≥10Hz Forensic Inference Log Rate

Teaming & Partnerships

Our team is structured as a prime contractor with deep autonomous systems software expertise, supported by a platform integration subcontractor with certified UAS hardware experience, a university AI assurance research partner providing independent algorithm validation, and an FFRDC partner providing independent test and evaluation support. We actively seek teaming discussions with prime integrators and platform OEMs pursuing DoD UAS autonomy programs.

TECHNICAL CAPABILITY AREAS

Capability Area	Description
Runtime AI Monitoring	Real-time behavioral envelope checking, confidence score auditing, and anomaly flagging via AI Mechanic onboard inference watchdog
Drift Detection	CUSUM/EWMA statistical monitoring of inference pattern drift across mission phases; configurable sensitivity

Capability Area	Description
Forensic Accountability	thresholds per platform/mission type Tamper-evident, cryptographically time-stamped inference logs; sub-second replay fidelity via Forensic Replay Module (FRM)
Supply Chain Integrity	SBOM/HSBOM verification against signed component manifests executed pre-mission; automated discrepancy alerting
Lifecycle Configuration Control	Cryptographically-signed model provenance ledger; Git-based version control; full traceability from training data to deployed inference artifact
Autonomous Intervention	Graduated ACE response ladder: alert → watchdog → graceful degradation → safe-state handover; <2-second full handover execution
GCS Integration	Real-time autonomy health dashboard via ASB ; ACE alert overlays; human override initiation; compatible with STANAG 4586 GCS architectures
Compliance Documentation	Automated Mission Readiness Reports (MRR) ; structured pathway support for MIL-HDBK-516C and DO-178C airworthiness certification

RELEVANT EXPERIENCE & ANALOGOUS PROGRAMS

- Delivered embedded AI health monitoring systems for Group 4 fixed-wing ISR platforms under DoD R&D contracts, achieving operational integration without airframe modification or mission profile disruption.
- Developed software configuration management frameworks for safety-critical avionics systems certified to **DO-178C**, including artifact generation and requirements traceability across full development lifecycle.
- Integrated autonomous decision-support tools with **STANAG 4586**-compliant ground control stations, delivering real-time telemetry visualization and operator alert interfaces under operationally representative conditions.
- Supported Army and Air Force airworthiness authority coordination for software-intensive autonomous systems, including development of airworthiness determination packages and safety case documentation aligned with Army Aviation Engineering Flight Activity (AEFA) and AFLCMC/WN requirements.

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